

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M. Farah Ahsan/192511434 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Farah Ahsan
Signature of the student:

35/10

✓

ASSESSMENT

TOOL-1

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Date: 21/06/26

Short answers:

1) Total data = 3GB

Request size = 3MB

Effective throughput = 120MB/s

Assume 1GB = 1000MB

a) Total number of request

$$3GB = 3000MB$$

$$\text{Number of requests} = \frac{3000}{3}$$

1000 requests

b) Total transmission Time

Convert 3GB to bits:

$$3GB = 3,000,000,000 \text{ bytes}$$

$$3,000,000,000 \times 8 = 24,000,000,000 \text{ bits}$$

Throughput:

$$120MB/s = 120,000,000 \text{ bits/second}$$

$$\text{Transmission time} = \frac{24,000,000,000}{120,000,000}$$

$$= 200 \text{ seconds}$$

Total request

1,000 requests

Transmission time

200 seconds

The Application layer provides services directly to users and applications. In FTP, it:

- * Provides file transfer services
- * Allows users to download and upload files
- * Support Authentication using usernames and Passwords
- * Manage file operations such as deleting, creating, listing of files
- * Uses Protocols such as FTP to provide organized communication between applications

8) Component

	Delay
Application layer	4ms
Transport layer	3ms
Network layer	5ms
Data link layer	2ms
Physical layer	1ms
Propagation delay	18ms
Transmission delay	12ms

1. Total Processing delay:

$$4+3+5+2+1=15\text{ms}$$

al end-to-end delay:

$$\text{Total delay} = \text{Processing delay} + \text{Transmission delay} + \text{Propagation delay}$$

$$= 15 + 12 + 18$$

$$= 45 \text{ ms}$$

Total end-to-end delay = 45 ms

- Application layer: 4ms - Provides service to the user such as file transfer, email or web communication
- Transport layer: 3ms - Segments data and provides reliability, flow control and error recovery.
- Network layer: 5ms - Adds logical addressing and determines the best path for the packet
- Data link layer: 2ms - Creates frames and handles error detection and delivery across a local link
- Physical layer: 1ms - Converts data into electrical, optical, or radio signals for transmission
- Transmission delay: 12ms - Time required to Pack bits onto the communication link
- Propagation delay: 18ms - Time taken by the signal to travel from the sender to the receiver

Therefore, the Packet experiences a total end-to-end delay of 45 ms

9) Useful data = 80MB

Total frame size = 1600 bytes

Protocol overhead per frame = 80 bytes

Data carried by each frame is

$$1600 - 80 = 1520 \text{ bytes}$$

Assume:

$$1 \text{ MB} = 1,000,000 \text{ bytes}$$

1. Total number of frames

$$80 \text{ MB} = 80,000,000 \text{ bytes}$$

$$\text{Frames} = \frac{80,000,000}{1520} = 52631.58$$

$$= 52,632 \text{ frames}$$

2. Total protocol transmitted

$$\text{Total overhead} = 52,632 \times 80$$

$$4,210,560 \text{ bytes}$$

$$= 4.21 \text{ MB}$$

3. Transmission efficiency:

$$\text{Efficiency} = \frac{\text{Useful data}}{\text{Total transmitted data}} \times 100$$

$$\text{Efficiency} = \frac{1520}{1600} \times 100 = 95\%$$

total overhead includes headers, trailers, addresses, sequence numbers and error checking information added by different OSI layers.

More overhead means more data must be transmitted

- It increases bandwidth usage
- It can increase transmission time
- In this, 80 bytes out of every 1600-byte frame are overhead so 95% of each frame carries useful data

⑩ Original medical record size = 240 MB

Compression = 30%.

Segment size = 1400 bytes

Data link overhead = 60 bytes per frame

1 MB = 1,000,000 bytes

a) Compressed file size:

30% Compression means remaining 70%.

$$240 \times 0.70 = 168 \text{ MB}$$

168,000,000 bytes

b) Total number of segments:

$$\text{Number of segments} = \frac{168,000,000}{1400} = 120,000 \text{ segments}$$

c) Total Protocol Overhead

$$= 120,000 \times 60$$

$$= 7,200,000 \text{ bytes}$$

Therefore

7.2 MB

Final amount of data transmitted across Physical layer

$$\{ \text{Total transmitted data} = \text{Compressed data} + \text{Protocol overhead} \}$$

$$= 168,000,000 + 7,200,000$$

$$= 175,200,000 \text{ bytes}$$

$$= 175.2 \text{ MB}$$

- Presentation layer: Compress medical records; provide encryption
- Transport layer: Divide the compressed data into segments
- Network layer: Provides logical addressing and selects path for the data
- Data link layer: Encapsulates segments into frames
- Physical layer: Converts frames into signals